

4 and 5-Axis additive manufacturing of parts represented using free-form 3D curves

Erkan Gunpinar^{*}, Serhat Cam

Istanbul Technical University, Inonu Caddesi, No:65, Gumussuyu, Istanbul, 34437, Turkey

ARTICLE INFO

Keywords:

Additive manufacturing
Free-form curves
Print-paths
4/5-axis printing
Accessibility

ABSTRACT

Layer-by-layer additive manufacturing is commonly utilized for additive manufacturing. Recent works utilize curved layers (rather than planar ones), on which print-paths are located, and outline their advantage over planar slicing. In this paper, free-form three-dimensional curves are utilized as input for the generation of print-paths, which covers the model to be printed and do not necessarily lie on either a planar or a curved layer. Such print-paths have been recently studied for 3-axis additive manufacturing, and a novel additive manufacturing process for the models represented using such curves are proposed for 4 and 5-axis additive manufacturing in this paper. The input curves are first subdivided into short sub-curves (i.e., segments), which are then merged to obtain print-paths with (collision-free) printing-head orientations along them. Thanks to additional two rotational axes of the printing-head, a less number of print-paths can potentially be obtained, which can reduce subdivisions in the input curves, and therefore, is desirable in additive manufacturing for improved mechanical properties in the printed parts. As a proof of concept, the print-paths with printing-head orientations along them are finally validated using an AM simulator and machine.

1. Introduction

Parts with complex geometries, which are impossible to obtain via subtractive manufacturing, can be built via additive manufacturing. Planar slicing is still a common approach in additive manufacturing to build parts layer-by-layer, which can increase the stair stepping effect in the printed model and can yield poor mechanical properties [1,2]. Singamneni et al. [3] suggested curved layer slicing for Fused Deposition Modeling (FDM) [4] and demonstrated that better mechanical characteristics have been observed for the parts printed using this approach compared to that printed via of planar slicing. Additionally, thanks to the additional rotations around two axes, 5-axis manufacturing systems is able to manufacture parts with remarkable improvements in their mechanical properties [2,5].

Ben et al. [6] introduced a method to manufacture parts represented with three-dimensional (3D) free-form curves. Print-paths are generated from these curves for 3-axis additive manufacturing. However, due to the accessibility of a print-head in 3-axis manufacturing (i.e., print-head collisions with previously deposited material), input curves should be subdivided into shorter ones that are considered as print-paths. This is, however, undesirable and leads to poor mechanical properties for the

printed parts. Fig. 1 shows a print-path having an arc shape with a tube (b, in green) around it (representing the deposited material along the print-path). The print-path should be subdivided into two separate print-paths (a, in green and blue) for collision-free 3-axis additive manufacturing. Otherwise, collisions happen (See the red cone representing bounding cone of a printing-head). Collision-free printing is possible if the printing-head starts deposition from the bottom of the green tube and material is then deposited along the blue tube from its bottom to top (a). On the other hand, 4 or 5-axis manufacturing enables printing it without any subdivisions thanks to the rotation ability of the printing-head (b, yellow cones). As a result, a method is introduced in this work to generate collision-free print-paths from free-form 3D curves for 4 or 5-axis additive manufacturing. With the best of our knowledge, there are no works in literature tackling the same accessibility problem.

This paper is organized as follows. Relevant prior art is mentioned in Section 2. Section 3 involves an overview and terminology of the proposed technique. Technical/algorithmic details of the method is elaborated in Sections 4 and 5. Experiments for validating the proposed method and their results will be discussed in Section 6. Finally, we conclude and recommend future works in Section 7.

^{*} Corresponding author.

E-mail address: gunpinar@itu.edu.tr (E. Gunpinar).

<https://doi.org/10.1016/j.gmod.2022.101137>

Received 22 November 2021; Received in revised form 24 February 2022; Accepted 20 March 2022

Available online 24 March 2022

1524-0703/© 2022 Elsevier Inc. All rights reserved.

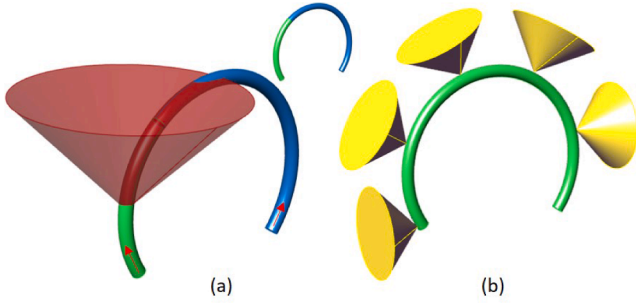


Fig. 1. (a) In 3-axis additive manufacturing, printing-head (the red cone representing bounding cone of a printing-head) starts depositing from the bottom of the green tube. The printing-head moves to the bottom of the blue tube and deposits material along it. Without printing-head retractions, collisions may happen between the printing-head and the previously deposited material (see the red cone). (b) Via 4 or 5-axis additive manufacturing, the printing-head (yellow cone) can rotate along axis/axes, and thereby depositing material along the print-path without requirement of subdivisions for the (input) print-path. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

2. Prior art

Tool accessibility and collision detection are important issues to be handled in manufacturing for successful production. Many papers have already focused on solving these problems in subtractive manufacturing. Tang et al. [7] proposed a sweep plane approach for global collision detection in 5-axis machining considering the change in workpiece geometry caused by material removal. Lartigue et al. [8] suggested a solution based on simulations of two different toolpaths and demonstrated the utility of this strategy in reducing surface faceting. Zhang et al. [9] used a model of cutter/workpiece engagement to examine tool orientation optimization in 5-axis ball-end milling. Sosin et al. [10] dealt with the approximation of general free-form surfaces for finding collision-free tangential cutting directions. For curvature-matched machining, Kim et al. [11] used hyper-osculating circles to approximate the machined surface. The toolpath was then optimized by maximizing the curvature-matching between tool and surface. Additionally, Barton et al. [12] used a toroidal cutter containing a negatively curved cutting area to achieve curvature adaption even for convex surfaces. Finally, Machchhar et al. [13] developed a system of algebraic constraints to model the tool sweep envelope using B-spline basis functions.

Additive manufacturing has drawn great attention in recent years due to its capability for building complex geometries. An analysis of the state of the art has been discussed by Livesu et al. [14]. Traditional 3-axis additive manufacturing utilizes planar slicing, which deposits material along a single direction and results in stair stepping effect and requirement of support structures [15,16]. As an obvious solution to these disadvantages, it is possible to enable material deposition in various directions by changing tilting angle for printing-head or work-piece. Chakraborty et al. [17] introduced the curved layers for FDM, in which the filaments were deposited along the paths on these layers. Xiao and Joshi [18] proposed an automated method to reorient the part via a five-axis machine during printing. Additionally, Isa and Lazoglu [19] developed another five-axis additive manufacturing machine, which was utilized for a model layered from its base to top. Dai et al. [20] presented a technique for fabricating 3D models on a robotic printing system equipped with multi-axis motion. Xu et al. [21] presented a new method for the 3 + 2-axis AM process planning, which eliminated the need for support by properly and continuously adjusting the build direction. Additionally, Chen et al. [22] utilized a variable-depth curved layer multi-axis additive manufacturing process for thin-shells.

Llewellyn-Jones [23] generated print-paths for curved layer fused filament fabrication process and demonstrated improvements in the

surface finish of the printed parts when using curved layers. Additionally, Etienne et al. [24] proposed a method involving slightly curved slicing compatible with a 3-axis printer for reducing support structures and staircase defects. Xu et al. [25] also produced curved layers (surfaces) from a mesh, on which print-paths were placed. Furthermore, Fang et al. [2] present a technique to design layers and print-paths to strengthen a model to be printed by aligning filaments along the directions with large stresses. Gunpinar and Armanfar [26] recently introduced a helical slicing method for multi-axis FDM, which allows users to adjust the lead angle and turn direction for the helical print-paths. Massarwi et al. [27] mapped geometries using curves for hierarchical tiling. Zhang et al. [28] proposed a motion planning method for supporting the manufacturing realization of the print-paths for multi-axis additive manufacturing. Mitropoulou et al. [29] presented a method for generating non-planar print paths of single-shell surfaces subject to the designer's input. Additionally, Gunpinar [30] recently generated curvature and feature-aware print-paths using hexahedral meshes, which can reduce stair step effect in the printed model. Finally, Ezair et al. [6] proposed an algorithm to form free-form curves from a 3D model of a part to be printed. The print-paths for 3-axis additive manufacturing were then ordered for avoiding collisions between printing-head and previously deposited material.

To the best of our knowledge, none of the works in literature have investigated and formulated the accessibility problem for free-form curves (representing the part to be printed) in 4 and 5-axis additive manufacturing. Thus, the work in this paper particularly contributes to this accessibility issue of free-form curves.

3. Overview and terminology

3.1. Overview

Given a three-dimensional model of a part (in orange) to be printed (Fig. 2 (a)), three-dimensional curves (b, in green) covering the model are obtained using the method proposed by Ben et al. [6]. The curves are then subdivided into shorter ones, which we call *segments* (c, in different colors). The print-paths are obtained next by merging consecutive segments. Red tubes (d) represent deposited material around print-paths. In 4/5-axis additive manufacturing, either the printing-head or printing-table rotates around two-axes. We assume that the printing-head rotates around X and Z -axes throughout this paper. The printing-head (illustrated with a yellow cone) orientations along the print-paths are determined for 4/5-axis additive manufacturing. The print-paths with printing-head orientations can finally be utilized for fabricating the part.

3.2. Terminology

Consider a set of parametric curves that covers the volume of an arbitrary 3D object. Each curve, $c(t)$, is uniformly subdivided into shorter segments, where $0 \leq t \leq 1$. All *segments* have a short length that is smaller than a predefined value, ϵ . Let $\mathcal{S}$ denote the set of all these segments, and let s be one segment, $s \in \mathcal{S}$. s has references to its previous and next segments that are denoted by s_{prev} and s_{next} , resp.

Printing orientation is expressed as a pair of angles (Φ, Θ) . Fig. 3 shows a printing-table with a printing-head and a printed object. To associate a direction with an orientation, a unit vector in the Z -direction is rotated around the X -axis by angle Φ , then around Z -axis by angle Θ . The resulting vector is the direction associated with the orientation (Φ, Θ) . The $\Phi \in [-\frac{\pi}{2}, \frac{\pi}{2}]$ and $\Theta \in [0, 2\pi]$ domains are divided uniformly in these domains by steps $\Delta\Phi$ and $\Delta\Theta$, resp. Let $(\Phi_0, \Phi_1, \dots, \Phi_M)$ and $(\Theta_0, \Theta_1, \dots, \Theta_N)$ represent the angles obtained after dividing these domains uniformly.

Definition 3.1. A printing orientation range, $\mathcal{PO}_{ij}$, is a set of

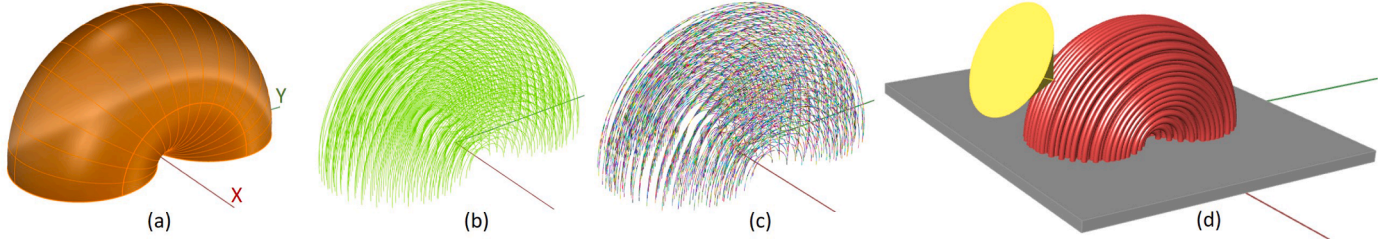


Fig. 2. Three-dimensional curves (b) are first obtained, which cover a three-dimensional model of a part (a, in orange) to be printed. Curves are then subdivided into short curves that is called segments (c, in different colors). Consecutive segments are merged to obtain print-paths (d, red tubes), and printing-head (yellow cone) orientations are computed, which are used to fabricate the part. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

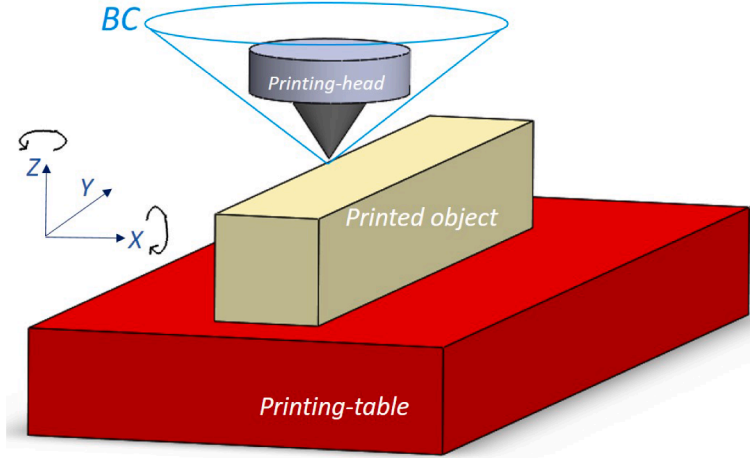


Fig. 3. A printing-head (in gray) over a printed-object (in yellow) can rotate around X and Z -axes (its bounding cone (BC) in blue). A printing-head over the printing-table (in red) can move along the X , Y and Z -axes. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

orientation ranges $([\Phi_i, \Phi_i + \Delta\Phi], [\Theta_j, \Theta_j + \Delta\Theta])$. Let BC be a bounding cone (of a printing-head) located at the origin with a cone angle, α , and a cone axis along the Z -direction. The expanded BC for $\mathcal{PO}_{ij}$, denoted $\mathcal{V}(\mathcal{PO}_{ij})$, is the minimal cone containing BC reoriented (rotated) in all orientations in $\mathcal{PO}_{ij}$.

$\mathcal{V}(\mathcal{PO}_{ij})$ denotes an expanded cone obtained when a bounding cone (BC, a blue cone in Fig. 3) of a printing-head is assumed to be at an initial orientation of (Φ_i, Θ_j) , and is expanded by rotating it around the X and then Z -axes by angles $\Delta\Phi$ and $\Delta\Theta$ (where $i \in [0, M-1]$ and $j \in [0, N-1]$),

resp. To do this, a BC is first formed at the origin with a cone axis along the Z -direction (See Fig. 4 (a)). The BC is then rotated around X -axis by Φ_i and $\Phi_i + \Delta\Phi$, and two other cones are obtained (See yellow and orange cones in Fig. 4 (b)). These cones are separately rotated around Z -axis by Θ_j and $\Theta_j + \Delta\Theta$, and four new cones (in cyan and blue) are generated (See Fig. 4 (c)). The expanded cone, $\mathcal{V}(\mathcal{PO}_{ij})$, is the cone bounding these four cones (See the red cone in Fig. 4 (d)), which also bounds all the cones generated for the BC orientations, $\{(\Phi, \Theta)\}$, when $\Phi_i \leq \Phi \leq \Phi_i + \Delta\Phi$ and $\Theta_j \leq \Theta \leq \Theta_j + \Delta\Theta$. The expanded cone bounding multiple cones is computed using Barequet and Elber's approach

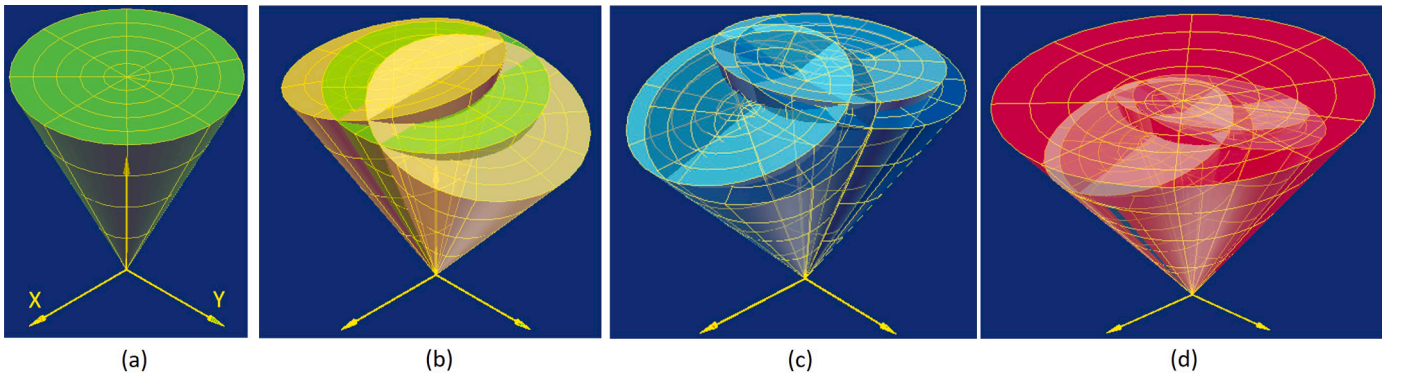


Fig. 4. The BC (with a cone axis along the Z -direction) is first placed at the origin (a) and rotated then around X -axis by $\Phi = -\frac{\pi}{12}$ and $\Phi + \Delta\Phi = \frac{\pi}{12}$, and two new cones (in yellow and orange) are generated (b). These cones are then separately rotated around Z -axis by $\Theta = -\frac{\pi}{12}$ and $\Theta + \Delta\Theta = \frac{\pi}{12}$, and four other new cones (in cyan and blue) are obtained (c). The expanded cone (in red) bounds these cones (d) and the cones generated for the BC orientations, $\{(\Phi, \Theta)\}$, when $-\frac{\pi}{12} \leq \Phi \leq \frac{\pi}{12}$ and $-\frac{\pi}{12} \leq \Theta \leq \frac{\pi}{12}$. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

[31].

In 4-axis additive manufacturing, the printing-head/printing-table rotates around only X-axis. Therefore, a printing orientation range, $\mathcal{PO}_i$, is defined as a set of orientation ranges $[\Phi_i, \Phi_i + \Delta\Phi]$ similar to Definition 3.1. The details of the proposed approach will be explained based on 5-axis additive manufacturing for the sake of generality.

Definition 3.2. The volume of a sweep of $\mathcal{V}(\mathcal{PO}_{ij})$ along a segment, s_m , in some orientation range, $\mathcal{PO}_{ij}$, is denoted by $\mathcal{V}(\mathcal{PO}_{ij}, s_m)$.

A segment is short and can be considered as a straight line. $\mathcal{V}(\mathcal{PO}_{ij})$ is located at an offset distance, z_{offs} , in the cone axis direction on the segments, allowing some space for the deposited materials. Fig. 5 illustrates two-dimensional view of $\mathcal{V}(\mathcal{PO}_{ij}, s_m)$ (in green) when a cone ($\mathcal{V}(\mathcal{PO}_{ij})$, in yellow) is swept along a segment (s_m , in black). Two cones are first placed at a distance, z_{offs} , from the start and end points of s_m . β , which is the angle between the cone axis, v , and segment direction, and γ are computed. Note that α , β and γ are on a straight line forming an angle of 180 degrees. The base center point (dark blue) of $\mathcal{V}(\mathcal{PO}_{ij}, s_m)$ is the point obtained by offsetting the center point of s_m . The Law of Sines is then employed to find the distance, h , between the apex point (light blue) of $\mathcal{V}(\mathcal{PO}_{ij}, s_m)$ and its base center point. As v' and v are in the same direction, the apex point can also be found. $\mathcal{V}(\mathcal{PO}_{ij}, s_m)$ is a truncated cone (in green) having apex and base center points. Fig. 5 (b) and (c) show two cones (in green and red) placed at an offset distance from the start and end points of the segment (in white) and a volume of sweep (in cyan) for a cone along the segment, resp. Finally, in case the cone axis and s_m are in the same direction, the cone bounding another is taken as $\mathcal{V}(\mathcal{PO}_{ij}, s_m)$.

Definition 3.3. A segment s_m is below a segment s_n , for $\mathcal{PO}_{ij}$, denoted $s_m(\mathcal{PO}_{ij}) < s_n(\mathcal{PO}_{ij})$, if there is an intersection between s_n and $\mathcal{V}(\mathcal{PO}_{ij}, s_m)$. Symmetrically, $s_m(\mathcal{PO}_{ij}) > s_n(\mathcal{PO}_{ij})$ denotes that s_m is above s_n for $\mathcal{PO}_{ij}$.

The term "below" here is metaphoric and useful to detect collisions between a printing-head and (already) printed segments. In other words, volume of a sweep, $\mathcal{V}(\mathcal{PO}_{ij}, s_m)$, for a cone (bounding the printing-head) generated while printing s_m (or moving along s_m) from $\mathcal{PO}_{ij}$ should never intersect with a segment, s_n , that is already printed. $s_m(\mathcal{PO}_{ij}) < s_n(\mathcal{PO}_{ij})$ means that s_m is below s_n for $\mathcal{PO}_{ij}$. Consequently, s_m can be printed from $\mathcal{PO}_{ij}$ iff there is no printed segment above it for $\mathcal{PO}_{ij}$.

Definition 3.4. Segments can hide one another from a printing orientation range (i.e., $s_m(\mathcal{PO}_{ij}) < s_n(\mathcal{PO}_{ij})$ and $s_n(\mathcal{PO}_{ij}) < s_m(\mathcal{PO}_{ij})$). We call such an arrangement a *cycle-of-invisibility*. In a cycle-of-invisibility, there can be h segments hiding each other, where $h \geq 2$.

Cycle-of-invisibility for $\mathcal{PO}_{ij}$ reflects the existence of segments,

which are impossible to be printed/ordered without collisions between printing-head and segments (s_m and s_n) in $\mathcal{PO}_{ij}$. Subdividing such segments can prevent cycle-of-invisibility [6].

4. Construction of graphs

4.1. Construction of accessibility graphs, $\{G(\mathcal{PO}_{ij})\}$

The ordered printing time steps are denoted by $\{t_k, k = 0, \dots, Q\}$. Here, t_0 denotes the initial/start printing time step, at which no segments are printed. Similarly, t_Q is the final time step, after which all (given) segments are printed. Accessibility for segments may change at each printing-time. More precisely, a segment accessible from $\mathcal{PO}_{ij}$, at t_k , can be inaccessible from $\mathcal{PO}_{ij}$, at t_{k+1} .

Definition 4.1. A segment, s_m , is accessible from $\mathcal{PO}_{ij}$, at a printing time step, t_k , if the following conditions are satisfied:

- There is no segment, printed above it for $\mathcal{PO}_{ij}$, before t_k .
- There is no segment belonging to a same curve (segments obtained after subdividing a same curve) with s_m and above s_m , at t_k .

Segments belonging to a same curve are potentially printed at a same printing time step, t_k , and therefore, one can hide another, at t_k . Furthermore, a segment can be inaccessible initially, at t_0 , due to a cycle-of-invisibility as explained in Definition 3.4. Fig. 6 illustrates this case, where an input curve has an ivy shape. Segments surrounded by the (blue) dotted rectangle are inaccessible from all printing-head orientations when 4-axis additive manufacturing is utilized. Therefore, in such cases, the input curves belonging to those segments should be subdivided until all segments are accessible from at least one direction, at t_0 .

A directed accessibility graph $G(\mathcal{PO}_{ij})$ is formed for a printing orientation range, $\mathcal{PO}_{ij}$ to figure out visibility relations between segments, as follows:

1. A vertex, V_s^{ij} , is assigned for each segment, $s_m \in \mathcal{S}$ in $G(\mathcal{PO}_{ij})$.
2. A directed edge from a vertex corresponding to s_m to that corresponding to a segment, s_n , is added in $G(\mathcal{PO}_{ij})$ if $s_m(\mathcal{PO}_{ij}) < s_n(\mathcal{PO}_{ij})$ and s_n is already printed/ordered.
3. A directed edge is also added in $G(\mathcal{PO}_{ij})$ from a vertex of s_m to that of s_n if s_m and s_n belonging to a same curve and $s_m(\mathcal{PO}_{ij}) < s_n(\mathcal{PO}_{ij})$.

Given a printing orientation range of the printing-head, $\mathcal{PO}_{ij}$, Algorithm 1 constructs $G(\mathcal{PO}_{ij})$. A graph vertex is added for any segment, s_m (Line 4). The Intersect function (Line 9) checks whether the swept volume, $\mathcal{V}(\mathcal{PO}_{ij}, s_m)$, intersects another segment, s_n . In case intersection exists and s_n is a printed segment or s_m and s_n belong to same curve (Line 9), a directed edge is added (Line 10) in $G(\mathcal{PO}_{ij})$ from the vertex, V_s^{ij}

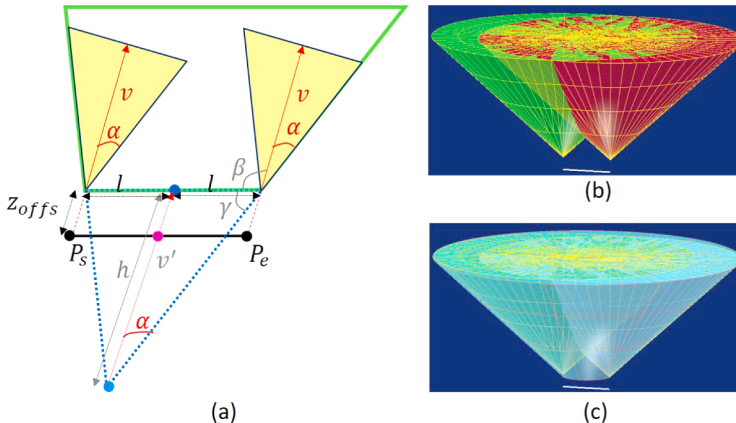


Fig. 5. (a) Volume of a sweep (in green) of a cone (in yellow) along a segment (in black) is a truncated cone with an apex (in light blue) and a base center (in dark blue) points. Volume of a sweep (c, in cyan) for a cone along a segment (in white) is a truncated cone bounding the cones (b, in green and red) that are placed at the start and end points of the segment (in white) from an offset distance in the cone axis direction. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

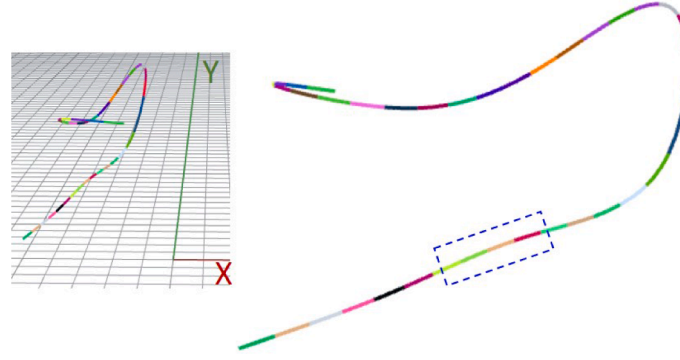


Fig. 6. Segments (in different colors) obtained after subdividing a curve can be initially inaccessible from all printing-head orientations (according to Definition 4.1) for 4-axis additive manufacturing, and therefore, the input curve should be subdivided until making all segments accessible from at least one orientation.

Input:

1. S , A set of segments;
2. $\mathcal{PO}_{ij}$, a printing orientation range;

Output:

1. $G(\mathcal{PO}_{ij})$, a directed accessibility graph;

Algorithm:

- 1: $G(\mathcal{PO}_{ij}) := \emptyset$;
- 2: **for all** $s_m \in S$ **do**
- 3: $V_{s_m}^{i,j} / V_{s_n}^{i,j} :=$ A vertex representing segment s_m/s_n in the printing orientation range $\mathcal{PO}_{ij}$;
- 4: $G(\mathcal{PO}_{ij}) := \text{AddVertex}(V_{s_m}^{i,j})$;
- 5: **end for**
- 6: **for all** $s_m \in S$ **do**
- 7: $\mathcal{V}(\mathcal{PO}_{ij}, s_m) := \mathcal{V}(\mathcal{PO}_{ij})$ swept along s_m ;
- 8: **for all** $s_n \in S$ **do**
- 9: **if** $\text{Intersect}(\mathcal{V}(\mathcal{PO}_{ij}, s_m), s_n) = \text{true}$ & $(\text{Printed}(s_n) = \text{true} \text{ or } \text{SameCurve}(s_m, s_n) = \text{true})$ **then**
- 10: $G(\mathcal{PO}_{ij}) := \text{AddDirectedEdge}(V_{s_m}^{i,j}, V_{s_n}^{i,j})$;
- 11: **end if**
- 12: **end for**
- 13: **end for**
- 14: **return** $G(\mathcal{PO}_{ij})$;

Algorithm 1. The construction of $G(\mathcal{PO}_{ij})$.

(corresponding to s_m) to $V_{s_n}^{i,j}$ (of s_n). Algorithm 1 is executed MN times, one for each $\mathcal{PO}_{ij}$ ($\forall i \in 0, \dots, N-1, \forall j \in 0, \dots, M-1$), to form MN directed graphs. Fig. 7 illustrates the segments (in red) having intersections with a volume of a sweep (blue cone) along a segment.

4.2. Construction of a root-segment graph, G_r

A root-segment graph is an undirected graph, and consists of vertices corresponding to only accessible segments and the edges between these vertices, if adjacent. Algorithm 2 outlines the construction of the root-segment graph, G_r , given $\mathcal{S}$ and $\{G(\mathcal{PO}_{ij})\}$. G_r is built using the vertices corresponding to the accessible segments of $\{G(\mathcal{PO}_{ij})\}$ as follows:

- A vertex $V_s^{i,j}$ is added in G_r (Line 5) if a segment, s , is accessible according to $G(\mathcal{PO}_{ij})$ (See the function Accessible in Lines 4, 6, 9 and 12).

- An edge is added between $V_s^{i,j}$ and $V_{s,prev}^{i,j}$ in G_r (Line 7) if segments, s and $s,prev$, are accessible according to $G(\mathcal{PO}_{ij})$ (i.e., adjacent segments, same $\mathcal{PO}_{ij}$).
- An edge is added between $V_s^{i-1,j}$ and $V_s^{i,j}$ in G_r (Line 10) if s is accessible according to $G(\mathcal{PO}_{(i-1)j})$ and $G(\mathcal{PO}_{ij})$ (i.e., same segment, adjacent $\mathcal{PO}_{ij}$).
- An edge is added between $V_s^{i,j-1}$ and $V_s^{i,j}$ in G_r (Line 13) if s is accessible according to $G(\mathcal{PO}_{(i-1)j})$ and $G(\mathcal{PO}_{ij})$ (i.e., same segments, adjacent $\mathcal{PO}_{ij}$).

5. Generation of print-paths

5.1. Selecting a print-path in G_r

A print-path, p , is a set of connected (accessible) segments in G_r , which is generated by merging adjacent segments (reaching them using the operators $s,prev$ and $s,next$). Orientation might change as printing-head moves from one segment to the next (moving between adjacent ranges of orientations). Multiple print-paths can be extracted from G_r ,

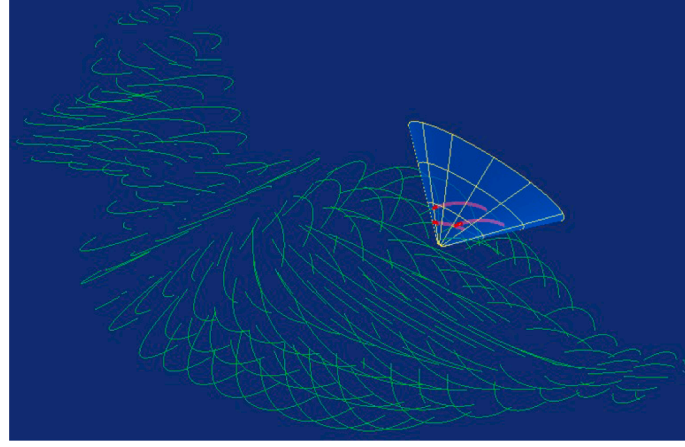


Fig. 7. A volume of a sweep (blue cone) for a cone along a segment is intersected with all segments to find the intersected segments (in red). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Input:

1. S , a set of segments;
2. $\{G(\mathcal{PO}_{ij})\}$, directed accessibility graphs for S , in all printing orientation ranges $\mathcal{PO}_{ij}$; $\forall i, j$;

Output:

1. G_r , a root-segment graph;

Algorithm:

```

1: for all  $s \in S$  do
2:   for  $i = 0$  to  $M - 1$  /* All  $\Phi$  intervals. */ do
3:     for  $j = 0$  to  $N - 1$  /* All  $\Theta$  intervals. */ do
4:       if Accessible( $G(\mathcal{PO}_{ij}), s$ ) = true then
5:         AddVertex( $G_r, V_s^{i,j}$ );
6:         if  $s_{prev} \neq \emptyset$  & Accessible( $G(\mathcal{PO}_{ij}), s_{prev}$ ) = true then
7:            $G_r := \text{AddEdge}(V_s^{i,j}, V_{s_{prev}}^{i,j})$ ;
8:         end if
9:         if  $i > 0$  & Accessible( $G(\mathcal{PO}_{(i-1)j}), s$ ) = true then
10:           $G_r = \text{AddEdge}(V_s^{i-1,j}, V_s^{i,j})$ ;
11:        end if
12:        if  $j > 0$  & Accessible( $G(\mathcal{PO}_{(i-1)(j-1)}), s$ ) = true then
13:           $G_r := \text{AddEdge}(V_s^{i,j-1}, V_s^{i,j})$ ;
14:        end if
15:      end if
16:    end for
17:  end for
18: end for
19: return  $G_r$ ;
  
```

Algorithm 2. Create root-segment graph (CreateRSGraph).

and therefore, the one based on any user-defined additive manufacturing criteria is selected.

Algorithm 3 summarizes the print-path selection method. All disconnected sub-graphs, $\mathcal{G}$, in G_r , are first found (Line 1). Graph vertices corresponding to segments belong to different curves are separated here. A source, r , and a target, t , vertices are separately added to each graph, $g \in \mathcal{G}$ (Line 3). Let $s_0, s_1, \dots, s_K$ be the segments (belong to a curve, c) having vertices in g , where K is integer. Graph edges are added in g from r and t to the node(s) belong to s_0 and s_K , resp. Vertical material deposition is desirable in additive manufacturing, and therefore, costs of the graph edges are set based on the angles between the

printing-head orientations (i.e., cone axes of BC) and segment. Lower costs are given to the graph edges having vertices corresponding to segments that have angles (with printing-head orientations) closer to 90 degrees. Dijkstra's algorithm [32] is then applied (Line 5) for finding the shortest path from r to t in each graph, g . Note that this path represents a print-path, p , with printing-head orientations, o , along p . p and o are then inserted (Line 6) to the lists, P and O , resp. The print-paths obtained are sorted next (Line 8) using one or more (user-defined) additive manufacturing criteria, which are suggested as follows:

Input:

1. G_r , a root-segment graph;

Output:

1. p , a print-path with o , printing-head orientations along p ;

Algorithm:

```

1:  $\mathcal{G} := \text{FindSubgraphs}(G_r)$ ;
2: for all  $g \in \mathcal{G}$  do
3:    $r, t := \text{AddSourceTargetNodes}(g)$ ;
4:    $g := \text{SetSourceTargetLinks}(r, t)$ ;
5:    $p, o := \text{Dijkstra}(r, t)$ ;
6:    $P, O := \text{Insert}(p, o)$ ;
7: end for
8:  $P, O := \text{Sort}(P, O)$ 
9: for all  $p \in P$  do
10:  if  $\text{Printability}(p) = \text{true}$  then
11:    return  $p, o$ ;
12:  else
13:     $p_1, o_1, p_2, o_2 := \text{Split}(p)$ ;
14:     $P, O := \text{Insert}(p_1, o_1, p_2, o_2)$ ;
15:  end if
16: end for
17: return  $\emptyset$ ;

```

Algorithm 3. Select a print-path in G_r (SelectPpath).

1. *Print ratio* (changing between 0 and 1) of p is $\frac{K+1}{M}$, where M is the number of segments (in c) that are not yet printed. Higher the print ratio, lower the subdivision is for c . Such subdivision is not preferable in AM as it leads to poor mechanical properties.
2. *Maximum Z* of p is the maximum (max) Z-coordinate along the segments, $s_0, s_1, \dots, s_K$. Minimum the max Z of p , print-paths closed to the printing-table can be selected (to a certain extent).

Priority is given for the first criterion during print-path selections. If print-ratio is same for multiple print-paths, the second criterion is utilized for the selections in this work. Let s_a and s_b be two segments that are not yet printed at a printing time step, t_k . Assume s_a be only accessible from a printing orientation, $\mathcal{PO}_{ij}$, and printing s_b , at t_k , result in hiding s_a , after t_k . Fig. 8 illustrates this case, in which there are three expanded bounding cones (BC_1, BC_2, BC_3) of a printing-head from different printing orientations (POs) and a set of segments (from the front view), resp. All segments are numbered from 1 to 9 and shown with different colors, at printing time step t_0 . Segment 1 is accessible from PO_2 , at t_0 , and printed at t_1 from this orientation. Segment 3 is accessible from PO_2 , after t_1 , and printed next, at t_2 , from PO_2 . Segment 6 is

accessible from PO_1 , after t_2 and printed, at t_3 , from PO_1 . Segments 2 and 5 are accessible, after t_3 , from PO_2 and PO_1 , resp. However, printing segment 5, at t_4 , from PO_1 hides segment 2 hence after (i.e., inaccessible forever) from all printing orientations. A possible solution is not to print segment 5 and print segment 2, at t_4 , using a printing orientation, PO_1 .

A *printability test* is required here for completely printing all segments covering the (input) 3D model. Given a print-path, p , accessibility for all unprinted segments is checked (Line 10 of Algorithm 3). In other words, unprinted segments should be accessible even after printing segments in p . This implies an assurance of accessibility of unprinted segments from at least a printing orientation all printing times. The test is conducted using each print-path starting from the beginning of P (sorted based on the above (user-defined) additive manufacturing criteria) until a print-path, p , of P passes this test (i.e., segments in p not hiding the unprinted segments forever). Algorithm 3 then returns p and o . Otherwise, p/o is split (Line 13) into two print-paths (with as much as equal sizes) and inserted (Line 14) in P/O to a sorted position according to the aforementioned additive manufacturing criteria.

5.2. Ordering/finding print-paths

Given a set of segments, $\mathcal{S}$, and accessibility graphs, $G(\mathcal{PO}_{ij})$, print-paths are computed/ordered, and printing-head orientations along them are computed as summarized in Algorithm 4. A root-segment graph, G_r , is first built (Line 3) via an algorithm explained in Section 4.2 (Algorithm 2). A print-path, p , with printing head orientations, o , along p is then chosen (Line 4), as described in Section 5.1 (Algorithm 3), and added to the lists, $\mathcal{P}$ and $\mathcal{O}$ (Line 5). Accessibility graphs are then updated (Line 6) while considering ordered segments in p , which are removed from $\mathcal{S}$ (Line 7). Lines 3–7 are executed until all segments in $\mathcal{S}$ are ordered. Fig. 9 (a) illustrates a print-path (in green) at a printing time step, along which orientations of a BC for a printing-head are depicted with cyan cones. It can be observed that BC orientation changes (b, red and blue cones) at the connection of two adjacent segments.

Lemma 5.1. Algorithm 4 always terminates with all segments in $\mathcal{S}$ printed, if $0 < \epsilon < z_{\text{offs}}$.

Proof. 1) Having $\epsilon > 0$, and assume $\epsilon < z_{\text{offs}}$. Consider a segment, s_a . Let z_0 be the lowest Z-coordinate of s_a . If the segment length is less than $\epsilon < z_{\text{offs}}$, the highest point in s_a is always lower than $z_0 + z_{\text{offs}}$ and therefore, below the printing-head. In other words, s_a can always be printed without printing-head collisions.

2) However, a segment, s_a , can be only accessible from a printing orientation, $\mathcal{PO}_{ij}$, at a printing time step, t_k , and hidden from $\mathcal{PO}_{ij}$, after t_k , due to printing another segment, s_b , at t_k . Thanks to printability test (Line 10 of Algorithm 3) over the unprinted segments, no segment will be inaccessible forever, and therefore, all segments can completely be printed. $\square$

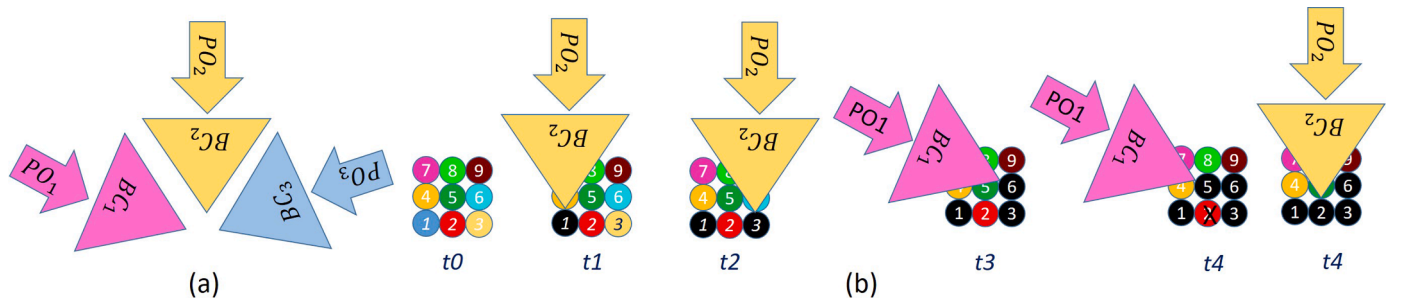


Fig. 8. (a) Three expanded bounding cones (BC_1, BC_2, BC_3) of a printing-head from different printing orientations (POs). (b) Segments are shown with different colors, at t_0 . At t_1, t_2 and t_3 , segments 1, 3 and 6 are printed from PO_2, PO_2 and PO_1 , resp. Printing segment 5 at t_4 , completely hides segment 2 so that segment 2 is not accessible from any POs. Thus, segment 2 is printed using PO_2 instead of printing segment 5 at t_4 .

Input:

1. $\mathcal{S}$, a set of segments;
2. $\{G(\mathcal{PO}_{ij})\}$, directed graphs for $\mathcal{S}$ in all orientations $\mathcal{PO}_{ij}$; $\forall i, j$;

Output:

1. $\mathcal{P}$, (ordered) print-paths, and $\mathcal{O}$, printing-head orientations along these print-paths;

Algorithm:

```

1:  $\mathcal{P}, \mathcal{O} := \emptyset$ ;
2: while  $\mathcal{S} \neq \emptyset$  do
3:    $G_r := \text{CreateRSGraph}(\mathcal{S}, \{G(\mathcal{PO}_{ij})\})$ ; /* Alg. 2 */
4:    $p, o := \text{SelectPpath}(G_r)$ ; /* Alg. 3 */
5:    $\mathcal{P}, \mathcal{O} := \text{AddPpath}(p, o)$ ;
6:    $\{G(\mathcal{PO}_{ij})\} := \text{UpdateAccessibilityGraphs}(p)$ ;
7:    $\mathcal{S} := \text{RemoveSegments}(p)$ ;
8: end while
9: return  $\mathcal{P}, \mathcal{O}$ ;

```

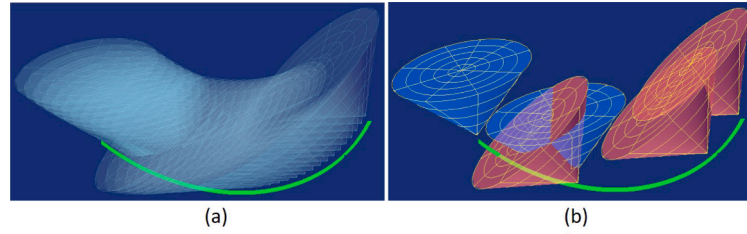
Algorithm 4. Generation of print-paths.

Fig. 9. (a) A print-path (in green) consists of segments, on which BCs (in cyan) of the printing-head are placed. (b) Near the middle of the print-path, BC orientation changes (See red and blue cones) at the connection of two adjacent segments. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

5.3. G-code generation for additive manufacturing

Before depositing material along the print-paths, reinsertion/retraction of the printing-head to/from the start/end of the print-paths should be carefully done to avoid possible collisions between the printing-head and previously ordered print-paths. The printing-head should be reinserted in/retracted from an orientation, (Φ, Θ) , from which the start/end segment of the print-path is also accessible. Let a /an start/end segment of the print-path is accessible from an orientation, $\mathcal{PO}_{ij}$. The printing-head should be reinserted in/retracted from an orientation that is a subset of $\mathcal{PO}_{ij}$.

G^1 continuities for the BC orientations of a printing-head at the connection points between adjacent segments should be assured to have a better quality for the manufactured parts. A BC orientation may change between adjacent segments as shown in Figs. 9 (c) and 10 (a). Fig. 10 (a) illustrates two adjacent segments (in green and blue) with their (two-dimensional) BCs (respectively, in orange and red). Let s_a and

s_b be two consecutive segments that can be accessible from $\mathcal{PO}_{ij}$ and $\mathcal{PO}_{i(j+1)}$, resp. We further subdivide s_a into smaller sub-segments and gradually change the BC orientations from its start, (Φ_i, Θ_j) , to end, $(\Phi_i, \Theta_j + \Delta\Theta)$. As s_b is also accessible from an orientation, $(\Phi_i, \Theta_j + \Delta\Theta)$, G^1 continuity for the BC orientation at the connection point of s_a and s_b can be achieved. Fig. 10 (a) illustrates two expanded BCs (in orange and red) along s_a and s_b . BC orientations (in cyan) along s_a changes smoothly (See Fig. 10 (b)) and the BC orientation at the end point of s_a is a subset of volume of sweep (in dark red) for BC along s_b . Finally, material deposition rate (radius) along the print-paths are adjusted in a way explained by Ben et al. [6].

6. Results and discussion

The proposed method was validated using several models represented using three-dimensional curves: two vertical semi-tori, two turbine blades, two vases, two ducks, a horizontal torus, a wavy and a

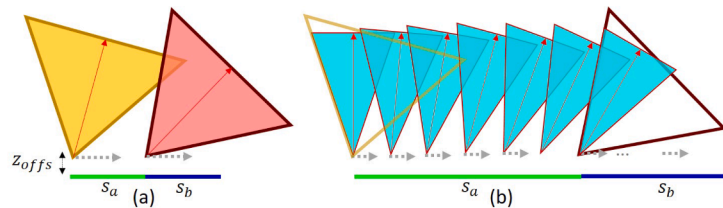


Fig. 10. (a) Expanded BCs (in orange and red) of two consecutive segments (s_a and s_b) represents the superset of BC orientations of a printing-head. (b) BC orientations (in cyan) change smoothly along s_a to guarantee the G^1 continuity of the BC orientation at the start point of s_b . (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

curved box. Cone angle, α , of a bounding cone (of the printing-head) is taken as $\frac{\pi}{2}$ radians. Additionally, $\Delta\Phi$ and $\Delta\Theta$ were set to $\frac{\pi}{6}$ radians in the 4 and 5-axis additive manufacturing experiments. z_{offs} , printing-head offset amount, was taken as the nominal material deposition radius, whose values change between 0.3 and 0.7, as shown in Table 1. To assure printing all segments (See Lemma 5.1), ϵ , segment length, should be smaller than z_{offs} .

Fig. 11 illustrates deposited material (red tubes) around print-paths and several BC (of a printing-head, yellow cones) orientations along print-paths. BC orientations change along the print-paths in such a way that vertical deposition (i.e., angles between BC orientations and segment directions are close to $\frac{\pi}{2}$ radians, Refer Section 5.1) can be achieved as much as possible (See Semi-torus-v1 (a), a semi-torus model vertically aligned). Semi-torus-h (i) is another semi-torus model distinctly horizontally oriented, whose print-paths are aligned in different isoparametric lines of the trivariate model compared to those in Semi-torus-v1 [6].

5-axis manufacturing was utilized for Semi-torus-h as many subdivisions of the input curves (covering the model) have been required to obtain collision-free print-paths in Semi-torus-h when using 4-axis manufacturing. Here, subdivision means different than subdividing an input curve into short segments that are merged to generate print-paths. Recall that the print-paths obtained can be formed either using whole (input) curve or some part of this curve. The latter case happens when (inevitably) subdividing an input curve for preventing cycle-of-invisibility (i.e., to ensure to print all given input curves). Such subdivision is undesirable and leads to poor mechanical properties for the printed parts [6].

Print-paths of Blade1 (c) and Blade2 (d) were also obtained for 4-axis manufacturing. It can be observed that the BC was tilted on the top of the models to achieve vertical deposition. For a vase (e, Twisted-vase1), 3-axis manufacturing can be utilized particularly for the bottom parts, while 4-axis manufacturing is required for the top parts as Φ was $\frac{2\pi}{3}$ radians there.

For printing a duck model (Duck2 (h)) using 4-axis additive manufacturing, we have replaced a different additive manufacturing criterion instead of the second one in Section 5.1, which is as follows:

- (1.) Minimum Z of p is the minimum (min) Z-coordinate along $s_0, s_1, \dots, s_K$. Minimum the min Z of p , print-paths closed to the printing-table can be selected (to a certain extent).

540 print-paths are obtained using this criterion (Min Z) in Duck2, while this number was 1142 for Duck1 (g), in which Max Z criterion (i.e., second criterion in Section 5.1) has been utilized. Thanks to the accessibility/root-segment graphs (Section 4) and printability test (described in Section 5.1), printing all segments was assured with

Table 1

Model properties and computational times of the algorithm steps (n_c : number of curves covering the model, n_s : number of segments after curve subdivision, z_{offs} : printing-head offset amount, T_a : computational time (in seconds) for computing accessibility graphs (Algorithm 1), T_p : computational time (in seconds) for generating print paths and printing-head orientations (Algorithm 4) and the time in the parenthesis is the processing time taken for the printability tests, T : total processing time for the proposed method).

Models	n_c	n_s	z_{offs}	T_a	T_p	T
Semi-torus-v1	262	16517	0.5	372	450 (301)	822
Semi-torus-v2	338	30423	0.35	1277	1578 (1119)	2855
Blade1	204	20447	0.3	554	929 (765)	1483
Blade2	255	5990	0.5	52	251 (225)	303
Twisted-vase1	223	3671	0.7	21	44 (29)	65
Twisted-vase2	790	33213	0.4	1441	8071 (7360)	9512
Duck1	515	6660	0.4	102	980 (870)	1082
Duck2	515	6660	0.4	104	525 (431)	629
Semi-torus-h	306	5445	0.5	177	150 (26)	327

collision-free extruder orientations. On the other hand, it has been observed that the subdivision amount of the input curves strongly depended on the heuristic utilized. Such subdivisions were less in case of Duck2.

6.1. Computational time

A PC with an Intel Core i7 6700 3.41 GHz processor and 16 GB memory was utilized in the experiments of this study. The implementation was single-threaded. The proposed algorithms were implemented in C# using Rhino - Grasshopper¹ library. Table 1 shows the number of curves, n_c , number of segments, n_s , and processing times taken in each algorithm steps for the models shown in Fig. 11. The total processing times changed from 65 to 9512 seconds. Higher the number of segments (as in Twisted-vase2), higher computational time was observed. Processing times for generating print paths and printing-head orientations along them (Algorithm 4) were mostly higher than those to compute accessibility graphs (Algorithm 1). Furthermore, printability tests dominated half of the processing times for most of the test cases. Finally, the computational times for Semi-torus-v2 and Twisted-vase2 were higher than Semi-torus-v1 and Twisted-vase1 as they were finely tessellated by the curves.

6.2. Virtual and physical additive manufacturing of models

Virtual additive manufacturing has been performed for Semi-torus-v1, Blade1, Twisted-vase1 and Duck2 using an AM simulator shown in Fig. 12 (a) (See video²). Furthermore, an AM machine³ (Fig. 12 (b)) was also utilized to manufacture the models (a wavy and a curved box) represented using print-paths (in different colors) shown in Fig. 13. The cone (in yellow) bounding the printing-head with its orientations along the print-paths can be seen in Fig. 13 (a) and (c). Printed parts are shown in Fig. 13 (d) and (e) for these models (See also manufacturing video⁴). The top-right image in Fig. 13 (d) depicts a printing-head orientation (during manufacturing), which is perpendicular to the printed part (at that printing time).

In real additive manufacturing, special care should be paid for the printing-head orientations to avoid collisions with the printing-table, which is shown with gray rectangular prism under the models in the Fig. 13 (a, c). An intersection check routine (between the printing-head and printing-table) is included in Algorithm 1 to avoid such collisions. Additionally, some constraints have arisen due to the nozzle geometry and heater block that we have used. Such constraints disallowed printing-head accessing the locations near to the printing-table (with certain angles). The apex angle of the BC (bounding the printing-head) was approximately 140°. This made manufacturing of highly-curved print-paths non-trivial. Finally, due to the B and C-axis rotational movements, the printing speed should be properly adjusted in manufacturing. Transition from linear to rotational motions (or vice versa) resulted in a change in the printing speed. This can affect the continuity of filaments, and therefore, a bad surface finish of the printed parts could be seen. To mitigate this problem, varying speeds for the rotational and linear motions have been utilized. The above-mentioned issues may lead to manufacturing accuracy errors between the input and manufactured models, which will be investigated in future studies.

6.3. Further discussions and limitations

Generation of support structures for 4 and 5-axis AM is more tricky compared to 3-axis AM as supports may be needed from any orientation

¹ <https://www.rhino3d.com/6/new/grasshopper/>

² <https://youtu.be/ITislexRJQ>

³ <http://www.5axismaker.com>

⁴ <https://youtu.be/2wZ0jZvP2sE>

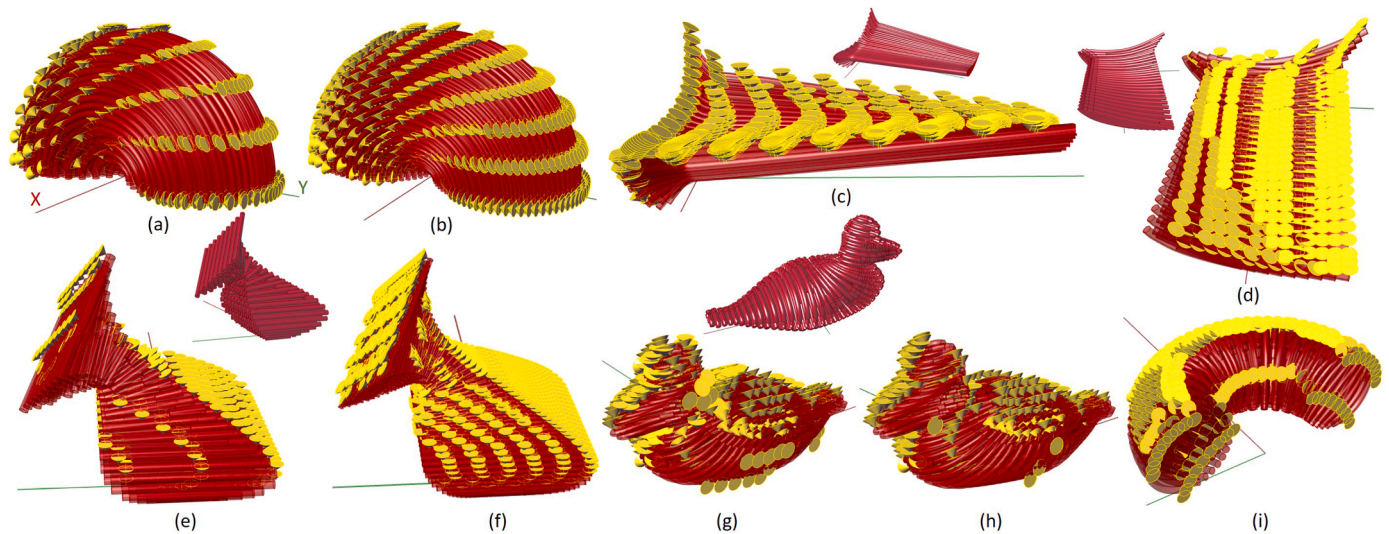


Fig. 11. Material deposited along print-paths are depicted with red tubes and yellow cones represent BC (of a printing-head) orientations along the print-paths for several models: Semi-torus-v1 (a), Semi-torus-v2 (b), Blade1 (c), Blade2 (d), Twisted-vase1 (e), Twisted-vase2 (f), Duck1 (g), Duck2 (h) and Semi-torus-h (i). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

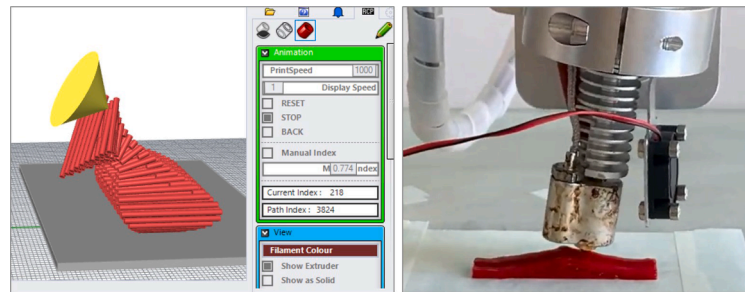


Fig. 12. Print-paths obtained from the proposed work have been tested using an additive manufacturing simulator (a) and 5AxisMaker 3D Printer, a multi-axis additive manufacturing machine (b). The printing-head is oriented in a way to deposit material during manufacturing perpendicular to the (currently) printed part as much as possible.

during printing. Basically, a support structure is added below a segment (to be printed) if there is no segment or base supporting (below) it in its (current) printing orientation. Therefore, supports should be created before those segments and extended down in the printing orientation until they hit the base or already printed segment(s). Support structure generation dedicated to 4 and 5-axis AM will be investigated further in a future study.

The present work considers/favors vertical deposition as deposited material not parallel to the surface/segment may lead to variable manufacturing conditions (i.e., asymmetric defects) [33]. Depending on the AM machine used gravity may also be considered as a constraint in such a way that depositing material along Z – direction can be identified as the favorable deposition direction. To achieve this, an alternative cost for the graph edges in $\mathcal{G}$ (Section 5.1) can be introduced, which is based on the angles between the printing orientation (to print a segment) and Z – direction.

There are two main collision detection problems in AM: local (collision between printing-head and manufactured model) and global (collision between printing-head and AM machine parts). In this work, both local and global (collisions with the printing-table) collisions have been taken into account as a 5-axis AM machine with a printing-table was utilized. While, for robotic AM [34], utilizing a cone bounding the printing-head may restrict the printing-head movements as whole robot body may be contained inside the cone (i.e., a global collision may occur). To prevent this problem, real printing-head geometry can be considered in return for higher computational cost (for intersection

checks between printing-head and segments) compared to utilizing a (simple) cone geometry.

7. Conclusion and future works

This paper introduces an approach for the generation of print-paths from free-form 3D curves covering the model to be printed in 4 and 5-Axis additive manufacturing. The curves are first divided into short sub-curves (segments) and print-paths are obtained via merging adjacent segments. The collision-free printing-head orientations along these print-paths are also computed. The printability test performed after selection of each print-path ensures printing all segments obtained after subdividing input curves.

The proposed algorithm is independent from the curve representation and takes any set of curves as input to generate (collision-free) print-paths. Curvature-aware print-paths recently proposed by Gunpinar [30] can also be given as input to the present work. As a future work, different heuristics for the selection of print-paths (Section 5.1) can be investigated, which potentially enable a less number of subdivision of input curves. Finally, the added flexibility and freedom of free-form curves can allow parts printed via 4/5-axis additive manufacturing with improved mechanical properties.

CRediT authorship contribution statement

Erkan Gunpinar: Conceptualization, Methodology, Software,

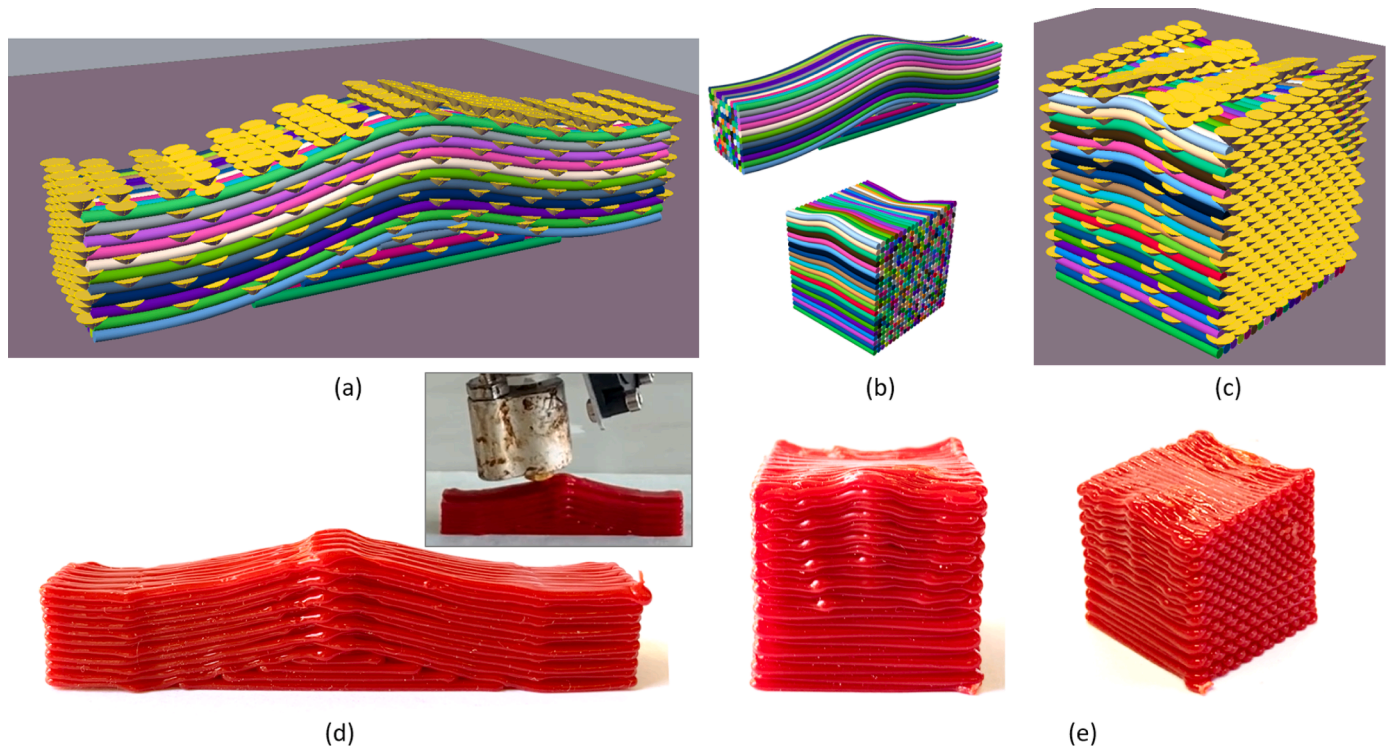


Fig. 13. A wavy (a) and a curved box (c) are represented using free-form curves (b, corresponding print-paths in different colors). The printing-head (in yellow) is oriented along the print-paths while considering vertical material deposition to the print-paths (top-right image in (d)) and avoiding collisions with printing-table (in gray)/already deposited material (a, c). Parts after additive manufacturing are shown in (d) and (e). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Funding acquisition, Writing – original draft. **Serhat Cam:** Methodology, Validation, Writing – original draft.

Declaration of Competing Interest

We wish to confirm that there are no known conflicts of interest associated with this publication and there has been no significant financial support for this work that could have influenced its outcome.

We confirm that the manuscript has been read and approved by all named authors and that there are no other persons who satisfied the criteria for authorship but are not listed. We further confirm that the order of authors listed in the manuscript has been approved by all of us.

We confirm that we have given due consideration to the protection of intellectual property associated with this work and that there are no impediments to publication, including the timing of publication, with respect to intellectual property. In so doing we confirm that we have followed the regulations of our institutions concerning intellectual property.

We understand that the Corresponding Author is the sole contact for the Editorial process (including Editorial Manager and direct communications with the office). He is responsible for communicating with the other authors about progress, submissions of revisions and final approval of proofs. We confirm that we have provided a current, correct email address which is accessible by the Corresponding Author.

Acknowledgments

The authors would like to thank Gershon Elber for his dedicated and valuable discussions having considerable impacts on the present work, and Arash Armanfar for helping in manufacturing of the parts. The research leading to these results has received funding from Scientific Research Center of Istanbul Technical University (Project number: 42541).

References

- [1] F. Alkadi, K.-C. Lee, A.H. Bashiri, J.-W. Choi, Conformal additive manufacturing using a direct-print process, *Addit. Manuf.* 32 (2020) 100975, <https://doi.org/10.1016/j.addma.2019.100975>.
- [2] G. Fang, T. Zhang, S. Zhong, X. Chen, Z. Zhong, C.C.L. Wang, Reinforced FDM: multi-axis filament alignment with controlled anisotropic strength, *ACM Trans. Graph.* 39 (6) (2020) 204, <https://doi.org/10.1145/3414685.3417834>.
- [3] S. Singamneni, A. Roychoudhury, O. Diegel, B. Huang, Modeling and evaluation of curved layer fused deposition, *J. Mater. Process. Technol.* 212 (1) (2012) 27–35, <https://doi.org/10.1016/j.jmatprotec.2011.08.001>.
- [4] C. Bellehumeur, L. Li, Q. Sun, P. Gu, Modeling of bond formation between polymer filaments in the fused deposition modeling process, *J. Manuf. Process.* 6 (2) (2004) 170–178, [https://doi.org/10.1016/S1526-6125\(04\)70071-7](https://doi.org/10.1016/S1526-6125(04)70071-7).
- [5] J. Gardner, T. Garabet, N. Kaill, R.I. Campbell, G. Bingham, D. Engstrom, N. Balci, Aligning material extrusion direction with mechanical stress via 5-axis tool paths. Bourell, D. (ed.). Proceedings of the 29th Annual International Solid Freeform Fabrication (SFF) Symposium An Additive Manufacturing Conference, 2018, pp. 2005–2019. <https://hdl.handle.net/2134/36347>. Austin, Texas, USA
- [6] B. Ezair, S. Fuhrmann, G. Elber, Volumetric covering print-paths for additive manufacturing of 3d models, *Comput.-Aided Des.* 100(2018) 1–13. [10.1016/j.cad.2018.02.006](https://doi.org/10.1016/j.cad.2018.02.006).
- [7] T.D. Tang, E.L.J. Bohez, P. Koomsap, The sweep plane algorithm for global collision detection with workpiece geometry update for five-axis NC machining, *Comput.-Aided Des.* 39 (11) (2007) 1012–1024, <https://doi.org/10.1016/j.cad.2007.06.004>.
- [8] C. Lartigue, C. Tournier, M. Ritou, D. Dumur, High-performance NC for high-speed machining by means of polynomial trajectories, *CIRP Ann. Manuf. Technol.* 53 (1) (2004) 317–320, [https://doi.org/10.1016/S0007-8506\(07\)60706-9](https://doi.org/10.1016/S0007-8506(07)60706-9). <https://hal.archives-ouvertes.fr/hal-00861903>
- [9] X. Zhang, J. Zhang, X. Zheng, B. Pang, W. Zhao, Tool orientation optimization of 5-axis ball-end milling based on an accurate cutter/workpiece engagement model, *CIRP J. Manuf. Sci. Technol.* 19 (2017) 106–116, <https://doi.org/10.1016/j.cirpj.2017.06.003>.
- [10] B. van Sosin, M. Bartoň, G. Elber, Accessibility for line-cutting in freeform surfaces, *Comput.-Aided Des.* 114 (2019) 202–214, <https://doi.org/10.1016/j.cad.2019.05.014>.
- [11] Y.-J. Kim, G. Elber, M. Bartoň, H. Pottmann, Precise gouging-free tool orientations for 5-axis CNC machining, *Comput.-Aided Des.* 58 (2015) 220–229, <https://doi.org/10.1016/j.cad.2014.08.010>.
- [12] M. Bartoň, M. Bizzarri, F. Rist, O. Sliusharenko, H. Pottmann, Geometry and tool motion planning for curvature adapted CNC machining, *ACM Trans. Graph.* 40 (4) (2021), <https://doi.org/10.1145/3450626.3459837>.

- [13] J. Machchhar, D. Plakhotnik, G. Elber, Precise algebraic-based swept volumes for arbitrary free-form shaped tools towards multi-axis CNC machining verification, *Comput.-Aided Des.* 90 (2017) 48–58, <https://doi.org/10.1016/j.cad.2017.05.015>.
- [14] M. Livesu, S. Ellero, J. Martinez, S. Lefebvre, M. Attene, From 3d models to 3d prints: an overview of the processing pipeline, *Comput. Graphics Forum* 36 (2) (2017) 537–564, <https://doi.org/10.1111/cgf.13147>.
- [15] S. Choi, J. Ryu, M. Lee, J. Cha, H. Kim, C. Song, D.-S. Kim, Support-free hollowing with spheroids and efficient 3d printing utilizing circular printing motions based on voronoi diagrams, *Addit. Manuf.* 35 (2020) 101254, <https://doi.org/10.1016/j.addma.2020.101254>.
- [16] G. Schuh, G. Bergweiler, G. Lukas, S. Hohenstein, J. Schenk, Feature-based print method for multi-axis material extrusion in additive manufacturing, *Procedia CIRP* 93 (2020) 85–89, <https://doi.org/10.1016/j.procir.2020.03.024>.
- [17] D. Chakraborty, B. Aneesh Reddy, A. Roy Choudhury, Extruder path generation for curved layer fused deposition modeling, *Comput.-Aided Des.* 40 (2) (2008) 235–243, <https://doi.org/10.1016/j.cad.2007.10.014>.
- [18] X. Xiao, S. Joshi, Process planning for five-axis support free additive manufacturing, *Addit. Manuf.* 36 (2020) 101569, <https://doi.org/10.1016/j.addma.2020.101569>.
- [19] M.A. Isa, I. Lazoglu, Five-axis additive manufacturing of freeform models through buildup of transition layers, *J. Manuf. Syst.* 50 (2019) 69–80, <https://doi.org/10.1016/j.jmsy.2018.12.002>.
- [20] C. Dai, C.C.L. Wang, C. Wu, S. Lefebvre, G. Fang, Y.-J. Liu, Support-free volume printing by multi-axis motion, *ACM Trans. Graph.* 37 (4) (2018), <https://doi.org/10.1145/3197517.3201342>.
- [21] K. Xu, L. Chen, K. Tang, Support-free layered process planning toward 3 + 2-axis additive manufacturing, *IEEE Trans. Autom. Sci. Eng.* 16 (2) (2019) 838–850, <https://doi.org/10.1109/TASE.2018.2867230>.
- [22] L. Chen, M.-F. Chung, Y. Tian, A. Joneja, K. Tang, Variable-depth curved layer fused deposition modeling of thin-shells, *Robot Comput Integr Manuf* 57 (2019) 422–434, <https://doi.org/10.1016/j.rcim.2018.12.016>.
- [23] T. Llewellyn-Jones, R. Allen, R. Trask, Curved layer fused filament fabrication using automated toolpath generation, *3D Printing and Additive Manufacturing* 3 (4) (2016) 236–243, <https://doi.org/10.1089/3dp.2016.0033>.
- [24] J. Etienne, N. Ray, D. Panozzo, S. Hornus, C.C.L. Wang, J. Martínez, S. McMains, M. Alexa, B. Wyvill, S. Lefebvre, Curvislicer: slightly curved slicing for 3-axis printers 38 (4) (2019), <https://doi.org/10.1145/3306346.3323022>.
- [25] K. Xu, Y. Li, L. Chen, K. Tang, Curved layer based process planning for multi-axis volume printing of freeform parts, *Comput.-Aided Des.* 114 (2019) 51–63, <https://doi.org/10.1016/j.cad.2019.05.007>.
- [26] E. Gunpinar, A. Armanfar, Helical5AM: five-axis parametrized helical additive manufacturing, *J. Mater. Process. Technol.* 304 (2022) 117565, <https://doi.org/10.1016/j.jmatprotec.2022.117565>.
- [27] F. Massarwi, J. Machchhar, P. Antolin, G. Elber, Hierarchical, random and bifurcation tiling with heterogeneity in micro-structures construction via functional composition, *Comput.-Aided Des.* 102 (2018) 148–159, <https://doi.org/10.1016/j.cad.2018.04.017>.
- [28] T. Zhang, X. Chen, G. Fang, Y. Tian, C.C.L. Wang, Singularity-aware motion planning for multi-axis additive manufacturing, *IEEE Rob. Autom. Lett.* 6 (4) (2021) 6172–6179, <https://doi.org/10.1109/LRA.2021.3091109>.
- [29] I. Mitropoulou, M. Bernhard, B. Dillenburger, Print paths key-framing: Design for non-planar layered robotic FDM printing, *Symposium on Computational Fabrication*, Association for Computing Machinery, New York, NY, USA, 2020, <https://doi.org/10.1145/3424630.3425408>.
- [30] E. Gunpinar, Curvature and feature-aware print-paths from hexahedral meshes for additive manufacturing, *Comput.-Aided Des.* 141 (2021) 103101, <https://doi.org/10.1016/j.cad.2021.103101>.
- [31] G. Barequet, G. Elber, Optimal bounding cones of vectors in three dimensions, *Inf. Process. Lett.* 93 (2) (2005) 83–89, <https://doi.org/10.1016/j.ipl.2004.09.019>.
- [32] J.A. Bondy, U.S.R. Murty, *Graph theory with applications*, Elsevier North-Holland, 1976.
- [33] M. Rauch, J.-Y. Hascoet, V. Querard, A multiaxis tool path generation approach for thin wall structures made with WAAM, *J. Manuf. Mater. Process.* 5 (4) (2021).
- [34] I.E. Yigit, I. Lazoglu, Spherical slicing method and its application on robotic additive manufacturing, *Prog. Addit. Manuf.* 5 (2020) 387394.